

CSCI 160 Homework 4

1. Simplify the Boolean function

$$F(A, B, C, D) = \sum m(0, 1, 2, 5, 8, 9, 10)$$

using a four-variable K-map. Then draw the NAND circuit for your simplified expression, **using the method we went over in class.**

2. Simplify the Boolean function

$$F(A, B, C, D) = \prod M(0, 1, 2, 3, 4, 8, 12)$$

using a four-variable K-map. Then draw the NOR circuit for your simplified expression, **using the method we went over in class.**

3. We want to check whether two 8-bit numbers $A = A_7A_6A_5A_4A_3A_2A_1A_0$ and $B = B_7B_6B_5B_4B_3B_2B_1B_0$ are equal. Using XNOR gates, write the Boolean expression for the output *Equal*, then use it to determine whether $A = 10110100$ and $B = 10010100$ are equal.
4. A 4-bit message $wxyz = 1011$ is sent using even parity. Find the parity bit P and write the full transmitted word $wxyzP$.